Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/08/2011
PI Name:	Nielsen, Sheila Ph.D.		
Project Title:	The fungal response to simulated microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sheila.nielsen@montana.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	406-994-5177
Organization Name:	Montana State University		
PI Address 1:	Leon Johnson Hall, room 315		
PI Address 2:			
PI Web Page:			
City:	Bozeman	**State:**	MT
Zip Code:	59717	**Congressional District:**	1
Comments:	PI name change to Sheila Nielsen in 2014 (formerly Sheila Nielsen-Preiss)--Ed., 1/12/2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	07/01/2009	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 6/30/2012 per A. Chu/ARC (Ed., 4/29/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hyman, Linda (Boston University)		
Grant/Contract No.:	NNX09AM79G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Modeling the microgravity environment, using rotating suspension culture bioreactors, allows investigation of eukaryotic cellular responses in a ground-based model and provides insights into the impact of space flight on cellular physiology. We have previously demonstrated that simulated microgravity induces morphologic changes and differential expression of several groups of yeast genes, including those related to morphogenic transformation, polarity determination, and the unique environmental stress of microgravity. The focus of this application is to characterize the molecular mechanisms responsible for the eukaryotic microbial response to microgravity using ground-based models. Choosing organisms from the fungal world is based on two criteria; (1) the microbial system is a powerful model to examine fundamental aspects of cell biology common to eukaryotic systems, such as polarity determination (Saccharomyces cerevisiae) and, (2) infection by yeast pathogens (Candida albicans) is a concern for human health. By using a comparative approach, we aim to identify responses that are conserved across yeast species and perhaps among higher eukaryotes. The overall objective of this proposal is to examine the hypothesis that cells exposed to microgravity undergo phenotypic and genotypic changes that alter polarity determination. The net result of altered polarity is a disrupted budding pattern in Saccharomyces cerevisiae and filamentation in Candida albicans consistent with increased pathogenicity. In Saccharomyces cerevisiae localization and expression levels of selected proteins (such as Rax1 and Bud2) will be evaluated in response to simulated microgravity using green fluorescence protein fusion constructs. The effects of simulated microgravity on growth and budding will also be evaluated in strains containing mutations in selected genes (including, but not limited to Cdc25, Bud5, Rax2). Signal transduction pathways involved in sensing mechanoreception driving these responses will be addressed. Ultimately, the propensity for increased biofilm formation and virulence will be evaluated in Candida albicans subjected to simulated microgravity. Given the flexible requirements for yeast growth and the availability of molecular tools in the yeast system, these ground-based studies can be readily adapted to space-flight testing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The overall objective of this proposal is to examine the hypothesis that cells exposed to microgravity undergo physical changes that affect the ability to sense direction, or have an altered polarity determination. One result of altered polarity in yeast is a changed budding pattern in Saccharomyces cerevisiae (Brewer's yeast) and filamentation in Candida albicans (an opportunistic pathogen) consistent with increased pathogenicity. In Saccharomyces cerevisiae the localization and expression levels of selected proteins will be evaluated in response to simulated microgravity. Inducible constructs will also be used to analyze the presence of changes in the nutrient microenvironment in low shear conditions. Ultimately, phenotypic markers associated with increased virulence will be evaluated in Candida albicans following exposure to low shear modeled microgravity. Our initial aim is to characterize the role of low shear modeled microgravity in S. cerevisiae gene expression and morphology. Progress on this aim has focused on the potential role of altered microenvironments produced secondary to the low shear bioreactor environment that may ultimately influence cellular metabolism. For this study, the plasmid 2DPT-G1(U) in S. cerevisiae strain D603 was used (in collaboration with Ross Carlson, PhD). The promoter-reporter plasmid (2DPT-G1(U)) expresses GFP under the control of the yeast GAL promoter, which is induced by galactose and repressed by glucose. If microenvironments do exist in LSMMG, it would be expected that nutrient diffusion to cells would be compromised, resulting in differential utilization of carbon sources compared to control conditions. This would be visualized as differential expression of the promoter-reporter plasmid. For these studies, cells were grown for 3 days in control or LSMMG conditions with complete medium (glucose), changing the medium daily. Since S. cerevisiae preferentially utilizes glucose rather than galactose, promoter-reporter expression provides information about carbon metabolism. Following three days of conditioning in HARV, media containing both glucose and galactose was added to the cells, which were grown for twenty hours in LSMMG or control conditions. Cells were then analyzed for GFP fluorescence using flow cytometry. There was in increased proportion of LSMMG-conditioned cells expressing GFP from the reporter construct, indicating a local absence of glucose and relatively high proportion of galactose (compared to control). These data can be explained by the creation of a glucose-depleted microenvironment due to limited nutrient diffusion based on the low shear environment. Cumulatively, this data suggests the presence of microenvironments in the HARV bioreactor under LSMMG. Analysis for other microenvironmental alterations are being explored. Our second aim is to characterize the role of low shear modeled microgravity in C. albicans gene expression and morphology associated with increased virulence. Initial studies evaluated whether signaling pathway(s) that perceive changes in gravitational/low shear forces subsequently regulate genes with a role in filamentation and biofilm formation. A mitogen activator protein kinase (MAPK) inhibitor consistently reduced temperature (37C)- and serum-induced filamentation. In addition, immunoblot analysis demonstrated serum-responsive phosphorylation of an extracellular signal-regulated kinase (ERK)-like protein (likely CEK1) which provides an additional tool to evaluate the activation of this signaling pathway during exposure to the low shear environment. Involvement of the ERK-like pathway will now be evaluated in the low shear bioreactor-induced C. albicans filamentation. Conversely, we have not been able to demonstrate a consistent effect of the protein kinase A inhibitor, at several concentrations, on C. albicans filamentation. We suspect this is due to a lack of cross-species reactivity and will, therefore, evaluate other aspects of this signaling pathway. In addition, we have spent a considerable amount of time this year evaluating the propensity to form biofilm communities upon exposure to low shear modeled microgravity. Although this analysis is ongoing, evaluation of several low shear modeled microgravity-induced biofilms has demonstrated a distinguishable morphology when compared to control bioreactor membranes. Cultures maintained in the bioreactor for 4-22 days, a timeframe during which filamentation is consistently observed, results in the formation of colonies on the membrane. While the biofilms cultured under conditions of low shear modeled microgravity are not consistently thicker than those cultured under control conditions, the organization of the biofilm communities are noticeably altered. Control membrane biofilms have a relatively uniform cellular distribution, whereas biofilms cultured under experimental conditions display a higher-ordered architecture with a greater range of cellular intensity - large voids within the biofilm, as well as areas of increased cell density. To quantify differences in cell density throughout the biofilm, the distribution of pixel-based staining intensity was measured using MetaMorph software. The control membrane biofilm has the most narrow range of intensity variation; whereas the low shear microgravity-induced biofilm has the broadest range of intensity variation. As part of these studies we also evaluated whether exposure to low shear modeled microgravity affect C. albicans

	antimicrobial resistance. Cultures were maintained in the bioreactor for 5, 8, and 12 days with medium being replaced at 24 hr intervals. At each timepoint, yeast numbers were normalized and grown in the presence of amphotericin B. The number of viable cells following drug exposure was determined by plated colony counts. C albicans exposed to conditions of microgravity consistently demonstrated an increased resistance to concentrations of amphotericin B up to 2 ug/ml. In addition, we observed an increase in colony morphology specific phenotypic switching over time. Gene expression data confirms that the cells have at least a 2-fold increase in Hwp1 expression when cultured in experimental conditions as compared to control. In summary, we have demonstrated that the low shear modeled microgravity environment is inducing phenotypic alterations in yeast that may be consistent with increased virulence. Initial evaluation of the microenvironment suggests there are alterations in nutrient availability, or waste product accumulation, that are contributing to a stress response. Future studies will characterize potential sources of stress that result in the observed phenotypic alterations.
Bibliography Type:	Description: (Last Updated: 06/23/2023)
Articles in Peer-reviewed Journals	Woolley CM, Nielsen-Preiss SM, Hyman LE. "The response of Candida albicans to simulated microgravity." Gravitational and Space Biology 2010 Aug;23(2):93-4. , Aug-2010